

Skills Summary

Two-Step Equations & Checking Solutions

Use this reference while completing your worksheet or when studying.

Skill 1 — Solve $ax + b = c$

Rule: undo in reverse order of operations — undo the added or subtracted number *first*, then undo the multiplication. Same move on both sides, every time.

Example: Solve $3x + 4 = 19$

Step 1. Undo the operations in reverse order: the $+ 4$ goes first, then the times 3.

Step 2. Subtract 4 from both sides, then divide by 3: $x = (19 - 4)/3 = 5$.

$$\Rightarrow x = 5$$

Try it: Solve $2x + 7 = 19$

check: $9 = 2x + 7$

Skill 2 — Solve $\frac{x}{a} + b = c$

Rule: same order — undo the added number first, then undo the division by multiplying both sides by a .

Example: Solve $\frac{x}{2} - 3 = 5$

Step 1. The subtraction is the outer operation, so add 3 to both sides first, then undo the division by 2.

Step 2. $x = (5 + 3) \cdot 2 = 16$.

$$\Rightarrow x = 16$$

Try it: Solve $\frac{x}{5} + 1 = 4$

check: $16 = x$

Skill 3 — A negative coefficient

Rule: treat $-ax$ like any coefficient: isolate it, then divide both sides by the *negative* number, sign included.

Example: Solve $9 - 2x = 1$

Step 1. Move the plain number away first, so the term with x stands alone before you divide by its negative coefficient.

Step 2. Subtract 9 from both sides: $-2x = 1 - 9$. Divide by -2 : $x = (1 - 9)/(-2) = 4$.

$$\Rightarrow x = 4$$

Try it: Solve $11 - 3x = 2$

check: $8 = 3x$

(!) Watch out: divide by the coefficient *with its sign*. Dividing $-2x = -8$ by 2 instead of -2 leaves the sign error in your answer.

Skill 4 — Check by substitution

Rule: a value is a solution only if substituting it makes the equation *true*: left side value = right side value. Checking catches sign and order-of-steps errors immediately.

Example: Is $x = 5$ a solution of $3x - 2 = 13$?

Step 1. Substitute the candidate value into the left side and compare the result with the right side of the equation.

Step 2. $3 \cdot 5 - 2 = \mathbf{13}$ — it equals the right side, so $x = 5$ *is* the solution.

Try it: Is $x = 3$ a solution of $5x + 1 = 20$? Evaluate $5x + 1$ at $x = 3$ and compare.

check 19 — not 20, so no